Real-Gas PVT Behaviour, Enthalpy, Entropy and Fugacity

from Cubic Equations of State: A Python Program

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1 What we set out to do

Real gases do not obey $PV_m = RT$, and at the pressures met in process equipment the deviation is large. The program takes a fluid, a temperature and a pressure, and returns the compressibility factor, the molar volume, the residual and total enthalpy and entropy, and the fugacity, from four cubic equations of state — van der Waals (VDW), Redlich–Kwong (RK), Soave–Redlich–Kwong (SRK) and Peng–Robinson (PR) — for 45 pure fluids. The theory follows Smith, Van Ness, Abbott and Swihart [1].

2 Theory used

2.1 One equation for all four models

Each model can be written in the single generic form

$$P = \frac{RT}{V_m - b} - \frac{a(T)}{(V_m + \epsilon b)(V_m + \sigma b)}, \qquad a = \Psi\,\alpha(T_r)\frac{R^2T_c^2}{P_c}, \qquad b = \Omega\,\frac{RT_c}{P_c}, \tag{1}$$

and only the constants in Table 1 change from one model to the next. The quantity $\mathrm{d}\ln\alpha/\mathrm{d}\ln T_r$ is needed later for H^R and S^R, so it is tabulated alongside.

Table 1: Model constants as coded in the program.

Model	ϵ	σ	Ω	Ψ	$\alpha(T_r)$	$\mathrm{d}\ln\alpha/\mathrm{d}\ln T_r$
VDW	0	0	1/8	27/64	1	0
RK	0	1	0.08664	0.42748	$T_r^{-1/2}$	$-1/2$
SRK	0	1	0.08664	0.42748	$[1+m(1-\sqrt{T_r})]^2$	$-m\sqrt{T_r}/[1+m(1-\sqrt{T_r})]$
PR	$1-\sqrt{2}$	$1+\sqrt{2}$	0.07780	0.45724	$[1+m(1-\sqrt{T_r})]^2$	$-m\sqrt{T_r}/[1+m(1-\sqrt{T_r})]$

For SRK, $m = 0.480 + 1.574\omega - 0.176\omega^2$; for PR, $m = 0.37464 + 1.54226\omega - 0.26992\omega^2$.

2.2 From pressure to Z

Substituting $Z = PV_m/RT$, $A = aP/(RT)^2$ and $\beta = bP/RT$ into Eq. (1) gives a cubic in Z:

$$Z^3 + c_2Z^2 + c_1Z + c_0 = 0, \qquad \begin{aligned} c_2 &= (\epsilon+\sigma-1)\beta - 1,\\ c_1 &= A + \epsilon\sigma\beta^2 - (\epsilon+\sigma)\beta(\beta+1),\\ c_0 &= -\big[A\beta + (\beta+1)\epsilon\sigma\beta^2\big]. \end{aligned} \tag{2}$$

Putting $(\epsilon,\sigma) = (0,1)$ recovers the familiar SRK form $Z^3 - Z^2 + (A - \beta - \beta^2)Z - A\beta = 0$, checking the general result.

2.3 Residual properties and fugacity

With $q = \Psi\alpha(T_r)/(\Omega T_r)$ and

$$I = \frac{1}{\sigma-\epsilon}\ln\frac{Z+\sigma\beta}{Z+\epsilon\beta} \qquad (\text{VDW: } I = \beta/Z), \tag{3}$$

the residual Gibbs energy is $G^R/RT = Z - 1 - \ln(Z-\beta) - qI$. Differentiating it with respect to temperature gives the two departure functions required, and the fugacity follows from $\ln\varphi = G^R/RT$ directly:

$$\frac{H^R}{RT} = Z - 1 + (\kappa - 1)\, qI, \qquad \frac{S^R}{R} = \ln(Z-\beta) + \kappa\, qI, \qquad \ln\varphi = Z - 1 - \ln(Z-\beta) - qI, \quad (4)$$

where $\kappa \equiv \mathrm{d}\ln\alpha/\mathrm{d}\ln T_r$ from Table 1 and $f = \varphi P$. Because all three descend from the same G^R, they are thermodynamically consistent with each other.

2.4 Absolute H and S

The ideal-gas part uses $C_P^{ig}/R = A + BT + CT^2 + DT^{-2}$ and a reference state of 298.15 K and 1 bar:

$$H^{ig} = \int_{T_{ref}}^{T} C_P^{ig}\,\mathrm{d}T, \qquad S^{ig} = \int_{T_{ref}}^{T} \frac{C_P^{ig}}{T}\,\mathrm{d}T - R\ln\frac{P}{P_{ref}}, \quad (5)$$

both integrated analytically, so $H = H^{ig} + H^R$ and $S = S^{ig} + S^R$ need no numerical quadrature.

3 How the program works

3.1 Solving the cubic

Rather than iterating, the program solves Eq. (2) in closed form by Cardano's method, then polishes each root with Newton's method on the original polynomial. Roots with $Z \le \beta$ are dropped: they would mean $V_m \le b$.

3.2 Choosing a root

If one root survives, that is the answer. If three survive, the largest is vapour-like and the smallest liquid-like; the middle root has $(\partial P/\partial V)_T > 0$, has no physical meaning, and is discarded. The program reports the root and its phase, but does not claim the two phases are in equilibrium — that needs the equal-fugacity condition of Section 5.3.

3.3 A precision trap at the critical point

At the critical point Eq. (2) has a *triple* root, so an error δ in the coefficients moves Z by about $\delta^{1/3}$. The five-figure Ω and Ψ of Table 1 return $Z_c = 0.3272$ for RK instead of the exact $1/3$ — an error of 6×10^{-3} from inputs good to 6×10^{-7}. The program stores $\Omega_{RK} = (2^{1/3} - 1)/3$ and $\Psi_{RK} = 1/[9(2^{1/3} - 1)]$ to full precision, after which every Z_c is right. Newton polishing does not help: the rounded equation genuinely has its root elsewhere.

3.4 Code layout

The engine is a package `pvt_eos` driven from `cubic_eos.py`, with separate modules for the constants and units, the 45-fluid database, the cubic and its solver (`eos.py`), the properties and fugacity (`properties.py`), saturation, and the command line. Fluid constants are not typed in by hand — a script parses Tables B.1 and C.1 of [1] straight out of the textbook PDF, so transcription errors are impossible; the 45 fluids are those appearing in both tables.

4 The interfaces

From the command line, `python cubic_eos.py point -s methane -e pr -T 200 -P 50` prints the EoS parameters, the volumetric results, the residual and absolute properties and the fugacity; `--all-eos` prints all four models side by side, `table` sweeps T or P into a CSV, `sat` gives the saturation state. Two graphical front ends are also provided: a Streamlit application, and the browser page of Fig. 1, which needs no installation and agrees with the Python engine to 1.3×10^{-15} relative.

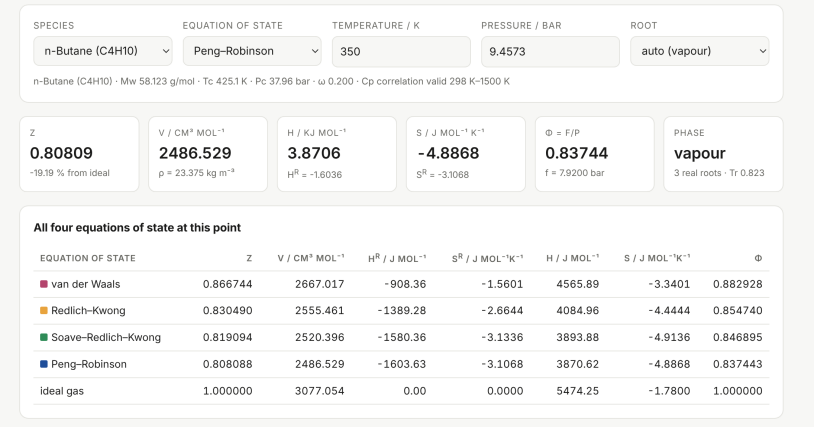


Figure 1: The browser interface, set to n-butane at 350 K and 9.4573 bar — the conditions of Example 3.9. The card row reports three real roots, and the molar volumes in the table reproduce the textbook values of Table 2.

5 Test runs

5.1 Single-phase case: CH_4 at 200 K, 50 bar

Here $T_r = 1.049$, $P_r = 1.087$, and every model returns a single root. All four give $Z < 1$ and $\varphi < 1$ — attraction still outweighs repulsion — but they spread: $Z = 0.5517$, 0.5523, 0.5529 and 0.5229 for VDW, RK, SRK and PR, with $\varphi = 0.7110$, 0.6921, 0.6923 and 0.6676. PR gives the densest fluid (173.9 against 183.5 cm^3/mol for VDW) and VDW the smallest departures ($H^R = -2.00$ against -2.53 kJ/mol for PR). With $H^{ig} = -3.124$ kJ/mol and $S^{ig} = -45.15$ J/(mol K), PR totals $H = -5.654$ kJ/mol and $S = -54.44$ J/(mol K).

5.2 Three roots: n-butane at 350 K, 9.4573 bar

This is Example 3.9 of [1], which quotes molar volumes for all four models, so it is the strongest available check. The program reproduces every tabulated value (Table 2; see also Fig. 1), with the intermediate quantities of the solution: $T_r = 0.8233$, $P_r = 0.2491$, $q = 6.6048$, $\beta = 0.02621$, $Z = 0.8305$. PR is closest to experiment on both branches, VDW furthest.

The same fluid at 500 K and 50 bar is Table 13.7 of [1], which tabulates Z, H^R and S^R for all four models, so it validates the departure functions and not only the volumetric part. Every entry is matched to within 0.1 %: RK gives $Z = 0.6852$, $H^R = -4503$ J/mol, $S^R = -6.542$ J/(mol K) against the printed 0.6850, -4505, -6.546, with $\beta = 0.09703$, $q = 3.8689$ and $I = 0.13247$ of Example 13.5 reproduced exactly.

Table 2: Saturated n-butane at 350 K and 9.4573 bar, in cm^3/mol, against Example 3.9 of [1].

	Experiment	VDW	RK	SRK	PR
V_V textbook	2482	2667	2555	2520	2486
V_V program	—	2667.0	2555.5	2520.4	2486.5
V_L textbook	115.0	191.0	133.3	127.8	112.6
V_L program	—	191.0	133.3	127.8	112.6

5.3 Saturation pressure from $f_L = f_V$

The pressure at which the two outer roots have equal fugacity is found by successive substitution on $P \leftarrow P \exp(\ln \varphi^l - \ln \varphi^v)$; the ideal-gas terms cancel, so $\Delta H^{vap} = H_V^R - H_L^R$. Over the seven benchmarks of Table 3 the mean absolute deviation is 1.01 % for PR and 2.05 % for SRK.

Table 3: Saturation states from equal fugacity.

Fluid	T [K]	P^{sat} exp. [bar]	SRK [bar]	dev. %	PR [bar]	dev. %	ΔH_{PR}^{vap} [kJ/mol]
H_2O	373.15	1.0133	0.9237	−8.85	0.9599	−5.27	42.10
H_2O	473.15	15.551	15.680	+0.83	15.575	+0.16	36.14
C_3H_8	300.00	9.9742	10.099	+1.25	9.9864	+0.12	14.75
C_4H_{10}	350.00	9.4573	9.5876	+1.38	9.4680	+0.11	17.47
CO_2	270.00	32.033	32.271	+0.74	31.899	−0.42	10.66
NH_3	300.00	10.613	10.625	+0.11	10.533	−0.76	20.41
C_2H_6	250.00	13.009	13.164	+1.19	13.038	+0.22	11.24

5.4 Consistency of the derived expressions

Equation (4) is also checked against independent numerical thermodynamics: across four fluids and all four models, H^R matches $-RT^2\,\mathrm{d}(G^R/RT)/\mathrm{d}T$ to 3.8×10^{-7} relative, S^R matches $-\mathrm{d}G^R/\mathrm{d}T$ to 4.7×10^{-7}, and $\ln\varphi$ matches $\int_0^P (Z-1)\,\mathrm{d}P/P$ by Simpson's rule to better than 2×10^{-4} — the only independent check on the fugacity, which no textbook table covers. As $P\to0$ it returns $Z\to1$, $\varphi\to1$ and zero residuals. In total 303 automated tests accompany the code.

6 What the results show

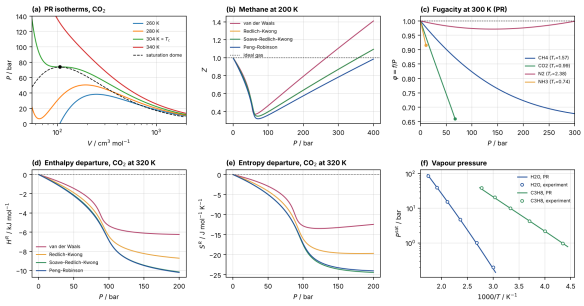


Figure 2: (a) PR isotherms for CO_2 with the saturation dome from $f_L = f_V$. (b) Z for methane at 200 K. (c) Fugacity coefficients at 300 K, each subcritical curve ending at its own saturation pressure. (d, e) Enthalpy and entropy departures for CO_2 at 320 K. (f) Vapour pressure against experiment.

Figure 2(a) shows what the cubic form exists to reproduce: one volume root above T_c, three inside the dome, a horizontal inflection at the critical point. The ranking in Fig. 2(b) — VDW < RK < SRK < PR — follows the critical compressibility each model predicts, 0.375, 0.333, 0.333 and 0.307 against a measured 0.286 for methane: VDW is 31 % high, PR only 7 %. The departures, Fig. 2(d, e), are negative throughout and grow with pressure, since attraction lowers the enthalpy and compression the entropy; all four agree at low pressure, where each reduces to the same second-virial limit. The fugacity coefficients, Fig. 2(c), show the same physics from the Gibbs-energy side, the departure growing as T_r falls.

7 Where the method falls short

- **Water.** At 373.15 K, PR gives $P^{sat} = 0.960$ bar against the steam-table 1.013 bar (4.8 % low), SRK 0.924 bar (8.8 % low): a two-parameter $\alpha(T_r)$ cannot describe hydrogen bonding.
- **The critical point.** Each model fixes Z_c at one universal number, while real values run from 0.229 (water) to 0.305 (hydrogen).
- **Input data.** Table C.1 of [1] prints $C_P^{ig}(298.15)/R = 10.720$ for 1,3-butadiene while the A, B, C coefficients on that row give 9.931; the program uses the coefficients. The fits hold only from 298 K to a stated T_{max}, so H and S below 298 K are extrapolated.
- **Pure fluids only.** Mixtures would need mixing rules for a and b, and partial fugacity coefficients.

Reference. [1] J. M. Smith, H. C. Van Ness, M. M. Abbott, M. T. Swihart, *Introduction to Chemical Engineering Thermodynamics*, 8th ed., McGraw-Hill. (Table 3.1; Tables B.1, C.1; Examples 3.9 and 13.5; Table 13.7.)